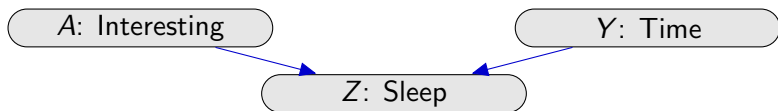


Discussion

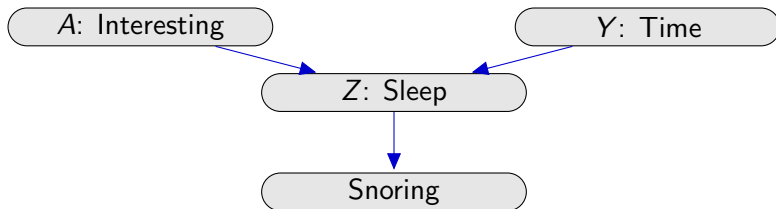
INFO/STSCI/ILRST 3900: Causal Inference

30 Sep 2026

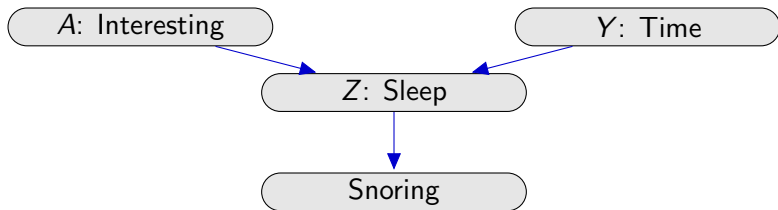
Collider



Collider



Collider



- If there is a causal path $X \rightarrow \dots \rightarrow Z$, then Z is a descendant of X

Colliders

For Colliders $\rightarrow Z \leftarrow$

Colliders

For Colliders $\rightarrow Z \leftarrow$

- ▶ If Z (or any descendant of Z) is in the conditioning set, then Z is open
- ▶ Otherwise Z is blocked

DAGs help us reason about exchangeability

DAGs help us reason about exchangeability

Procedure

- 1) List all paths between A to Y
- 2) Cross out the blocked paths (conditional on L)
- 3) Conditional Exchangeability holds if all remaining paths are causal

Open or blocked?

How to check if a path is open or blocked:

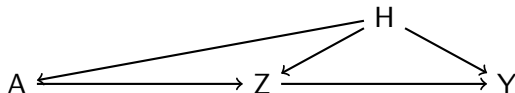
1. Traverse the path node by node
2. If any node is blocked, the entire path is blocked
3. If all nodes are open, then entire path is open

How to check if a node is open or blocked given a conditioning set:

- ▶ If non-collider:
 - ▶ Open if it is not in the conditioning set
 - ▶ Blocked if it is in the conditioning set
- ▶ If collider:
 - ▶ Open if it or any of its descendants are in the conditioning set
 - ▶ Otherwise it is blocked

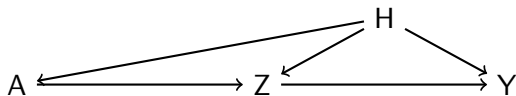
In-Class Assignment

Work through the DAG on your own, then we will solve it together

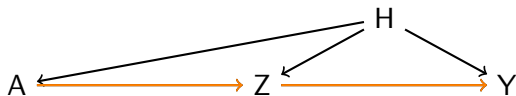


- ▶ What are the paths from A to Y?
- ▶ Determine if each of the paths is causal or non-causal
- ▶ Determine whether each node on each path is a collider or non-collider

Practice



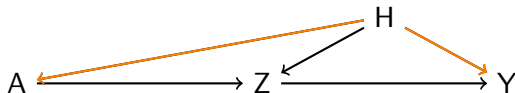
Practice



► $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

causal path

Practice



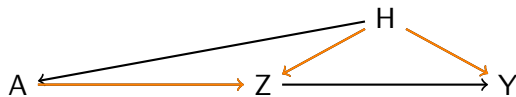
$$\blacktriangleright A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$$

causal path

$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

non-causal

Practice



$$\blacktriangleright A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$$

causal path

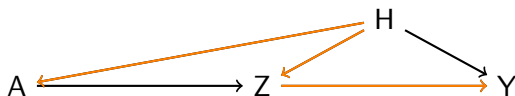
$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

non-causal

$$\blacktriangleright A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

non-causal

Practice



► $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

causal path

► $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$

non-causal

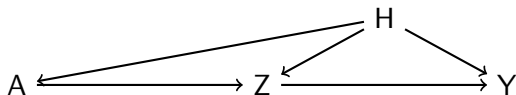
► $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$

non-causal

► $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

non-causal

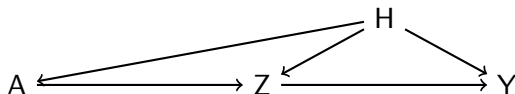
Practice



If we condition on $L = \emptyset$, which paths are open? Which paths are blocked?

- ▶ $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$
- ▶ $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

Practice



If we condition on $L = \emptyset$, which paths are open? Which paths are blocked?

$$\blacktriangleright A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$$

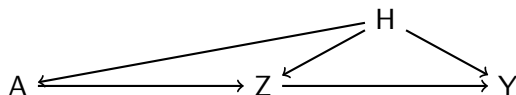
Open

$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

$$\blacktriangleright A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$$

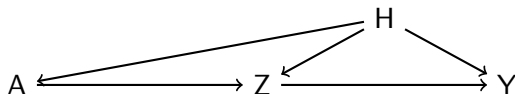
Practice



If we condition on $L = \emptyset$, which paths are open? Which paths are blocked?

- ▶ $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

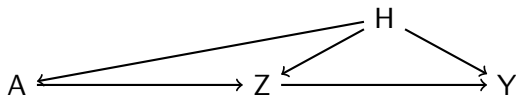
Practice



If we condition on $L = \emptyset$, which paths are open? Which paths are blocked?

- ▶ $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Blocked
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

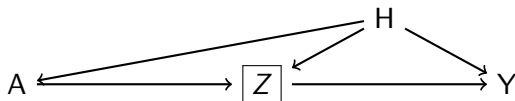
Practice



If we condition on $L = \emptyset$, which paths are open? Which paths are blocked?

- ▶ $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Blocked
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ Open

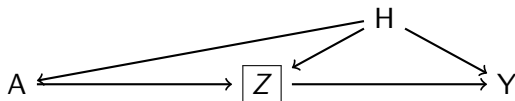
Practice



If we condition on $L = \{Z\}$, which paths are open? Which paths are blocked?

- ▶ $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$
- ▶ $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

Practice



If we condition on $L = \{Z\}$, which paths are open? Which paths are blocked?

$$\blacktriangleright A \rightarrow \underbrace{\boxed{Z}}_{\text{NC}} \rightarrow Y$$

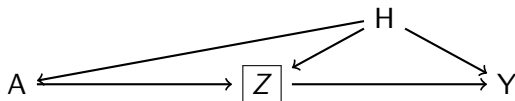
Blocked

$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

$$\blacktriangleright A \rightarrow \underbrace{\boxed{Z}}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{\boxed{Z}}_{\text{NC}} \rightarrow Y$$

Practice



If we condition on $L = \{Z\}$, which paths are open? Which paths are blocked?

$$\blacktriangleright A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$$

Blocked

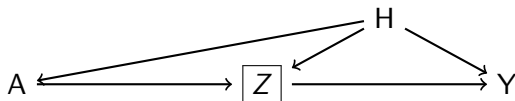
$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

Open

$$\blacktriangleright A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$$

$$\blacktriangleright A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$$

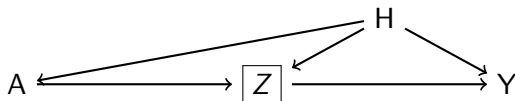
Practice



If we condition on $L = \{Z\}$, which paths are open? Which paths are blocked?

- $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ Blocked
- $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Open
- $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Open
- $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$

Practice



If we condition on $L = \{Z\}$, which paths are open? Which paths are blocked?

- ▶ $A \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ Blocked
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \rightarrow \underbrace{Z}_{\text{Col}} \leftarrow \underbrace{H}_{\text{NC}} \rightarrow Y$ Open
- ▶ $A \leftarrow \underbrace{H}_{\text{NC}} \rightarrow \underbrace{Z}_{\text{NC}} \rightarrow Y$ Blocked